# Jaineel Modi

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## Education

**Queen's University** **Expected May 2028**

*Bachelor of Computing (Honours) - Specialization in AI, GPA: 3.75/4.3, Dean's Honour List x3* *Kingston, ON*

- **Courses**: Data Structures & Algorithms, Software Architecture, Database Management Systems, Artificial Intelligence
- **Activities**: HackTheNorth (University of Waterloo), KingsHacks, AI Society, Competitive Programming Club

## Experience

**Shopify** **Incoming Winter 2027**

*Software Developer Intern* *Toronto, ON*

- Joining Shopify's engineering team to build e-commerce production software for millions of merchants.

**Royal Bank of Canada (RBC)** **May 2026 – Dec 2026**

*Software Developer Intern* *Toronto, ON*

- Architected the persistence service and **REST API** for the mortgage Application Lifecycle Management system, with one-fetch reads, indexed and capped queries, optimistic locking, and safe write ordering at **15K+** requests a week.
- Delivered a **Java Spring Boot** microservice with **Kafka** event-driven architecture that automated mortgage document validation against **581** business rules with exact and fuzzy matching, cutting manual review by **95%**.
- Modernized the configuration system used by **50+** mortgage teams across RBC on **MongoDB**, storing **10K+** records.

**Mulvey & Banani International** **Jun – Aug 2024 & May – Aug 2025**

*Software Developer Intern* *Toronto, ON*

- Launched a Revit Copilot powered by **GPT-4** via the **OpenAI API**, automating **20+** building-design (BIM) tasks from natural-language prompts with multi-step tool execution, serving **80+** users and handling **5K+** prompts per month.
- Built a Revit plugin (**Revit API**, **C#**, **.NET**) that auto-places **250+** electrical parts, saving **26 hours** per project.
- Implemented a WPF tool with **Python** and **FastAPI** to manage files across projects, used daily by **200+** employees.

**ConvoSearch** **Jan 2024 – May 2024**

*Software Developer Intern* *San Francisco, CA*

- Engineered an AI shopping interface with **GPT-4** and **LangChain** (few-shot, Chain-of-Thought) validating client data in an ETL pipeline at **98.3%** accuracy, with reranking lifting conversion **18%** and order value **25%**.
- Created **Python** and **MongoDB** CRUD APIs with indexing and pagination for **50K+** records, cutting runtime **30%**.

## Research

**RBC Borealis (LSi)** **Oct 2026 – Dec 2026**

*Student Researcher* *Toronto, ON*

- Leveraged **computer vision** and tabular ML models for anemia detection in 4 countries under **Dr. Eirene S.**
- Reached **0.92 AUROC** for detecting anemia in Ghanaian children with **TabPFN** on 29+ color features in **PyTorch**.

**Queen's University (RISE Lab)** **Sep 2025 – Jan 2026**

*Student Researcher* *Kingston, ON*

- Researched failure tracing in multi-agent LLM systems under **Prof. Yuan Tian**, pinpointing faulty agents and steps.
- Lifted **Qwen2.5-Coder** pass@1 on **HumanEval** by **12.8 pts** by building a **PyTorch** pipeline that logs multi-agent trajectories and trains a tracer with counterfactual replay and fault injection.

## Projects

**Vision** | *Next.js, TypeScript, Tailwind, Three.js, Gemini API, ElevenLabs* |

- Prototyped an AI pitch-coaching platform with simulated investors that reached **HackTheNorth 2026** final round.
- Architected a deliberation engine where **11** LLM agents role-play personas drawn from **326+** profiles across **25+** VC firms, cross-examining voice pitches and voting via a **Gemini** layer to give founders a weighted investment verdict.

**AudioMark** | *Flutter, Dart, just_audio, Provider* |

- Co-built a **Flutter** music player for dancers (section looping, speed control) live on **Google Play** with **5K+** installs.
- Managed the SDLC from design to **Google Play** release, with automated unit and E2E tests via **GitHub Actions**.

## Technical Skills

**Languages**: Python, Java, C, C++, C#, TypeScript, JavaScript, Dart, Ruby, SQL, Bash
**Frameworks & DB**: Spring Boot, React, Next.js, Node.js, Flask, .NET Core, PostgreSQL, MySQL, MongoDB
**AI & ML**: Claude Code, OpenAI API, LangChain, scikit-learn, Pandas, Hugging Face, PyTorch, TensorFlow, ONNX
**Tools**: AWS, Azure, Docker, Kubernetes, Git, Kafka, REST APIs, CI/CD, Microservices, ETL, LLM Agents